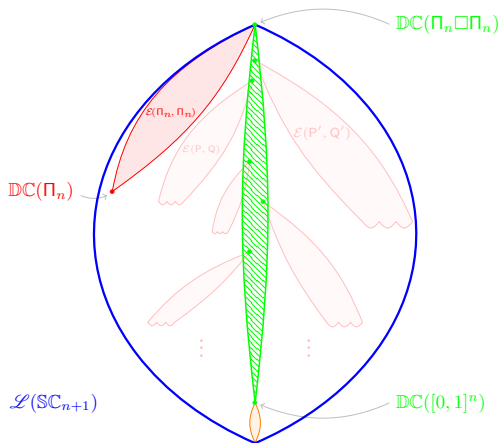


# How many generalized permutahedra are there?

**Germain Poullot** with Georg Loho & Arnau Padrol



24 March 2026, Combinatorial CoWorkspace  
arXiv:2510.03177

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Yes, but let's try to get a slightly more precise answer, shall we?

## 1 Generalized permutohedra, submodular cone

- 4 definitions
- Deformations

## 2 GP-sums and equality sets

- $\text{SC}_{n+1} \simeq \text{SC}_n^2$  more or less
- Drawing  $\mathcal{L}(\text{SC}_n)^2$  inside  $\mathcal{L}(\text{SC}_{n+1})$

## 3 Applications (now, we can count stuffs)

- Rays
- $f$ -vector, total number of faces, non-simplicial faces

# *Generalized permutohedra, submodular cone*

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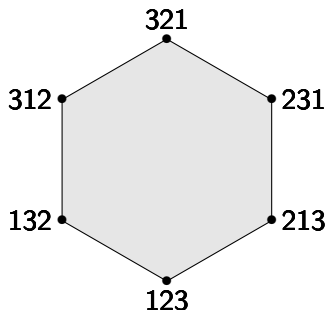
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## Example (Permutahedron)

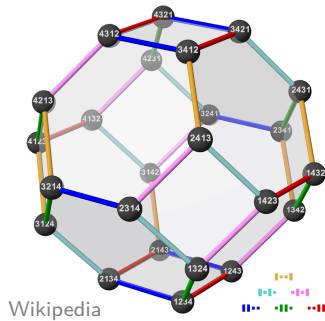
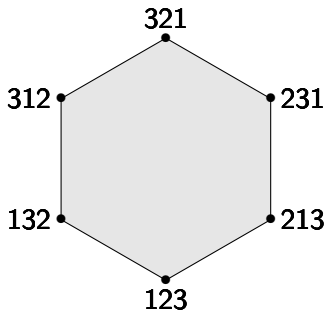
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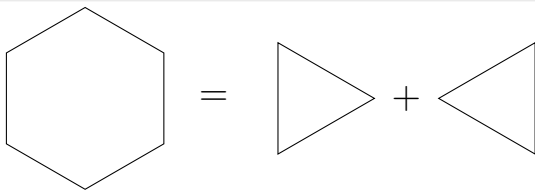
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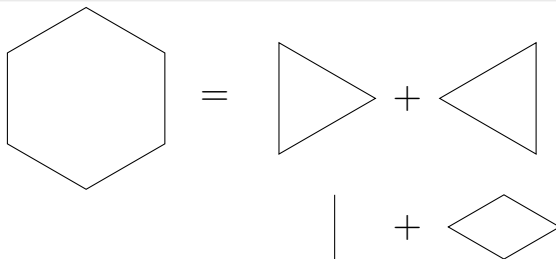
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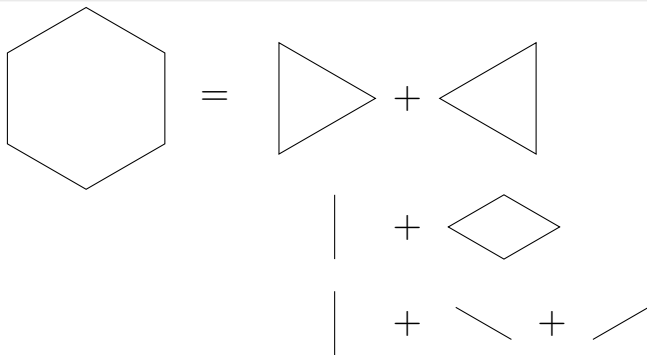
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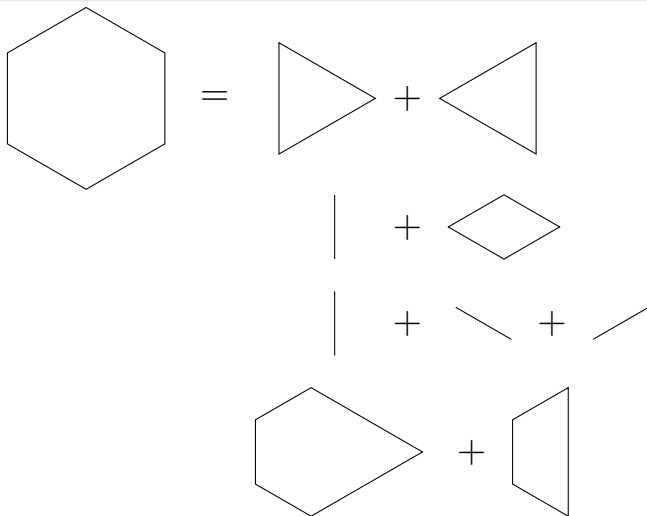
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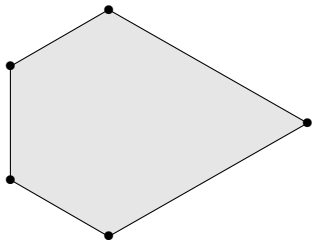
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$\mathcal{N}_P$ : Normal fan of polytope  $P$

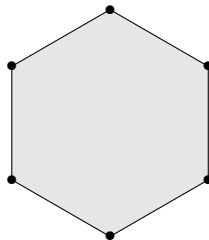
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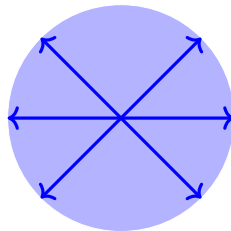
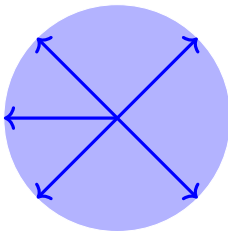
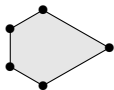
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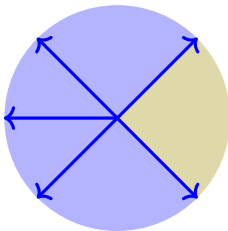
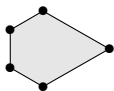
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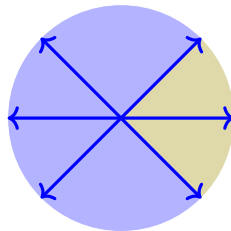
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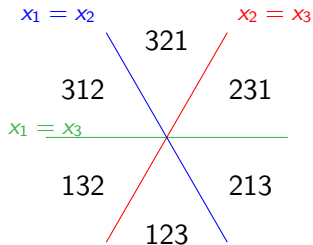
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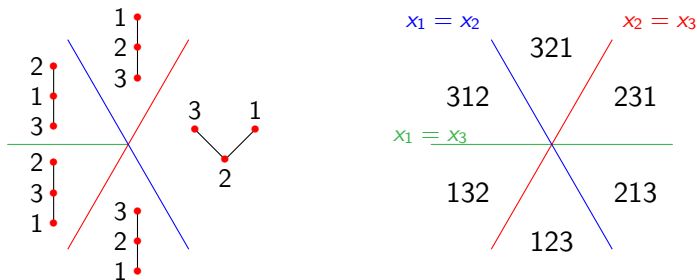
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- Set functions  $f : 2^{[n]} \setminus \{\emptyset, [n]\} \rightarrow \mathbb{R}$  which are *submodular*, i.e.

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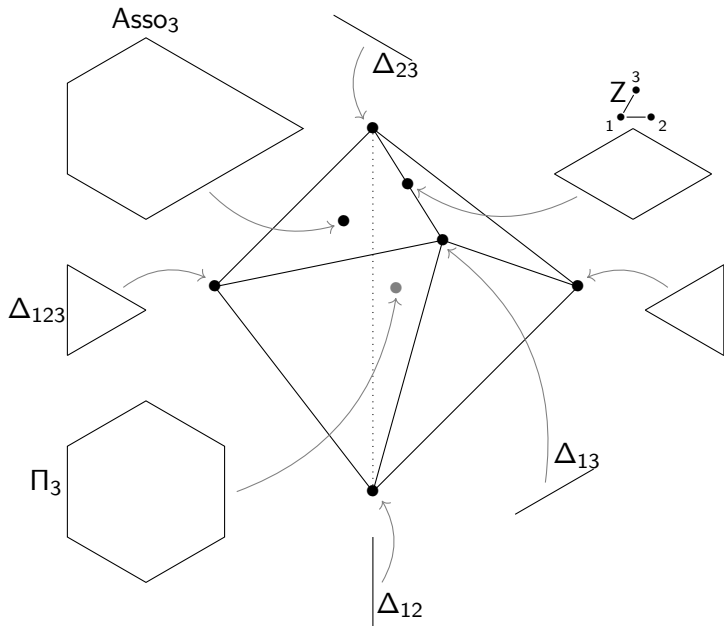
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*Submodular cone*  $\text{SC}_n := \{\text{all generalized } n\text{-permutohedra}\}$



$$P, P' \in \mathbb{SC}_n$$

$P'$  is a *deformation* of  $P$ , denoted  $P' \trianglelefteq P$ , iff:

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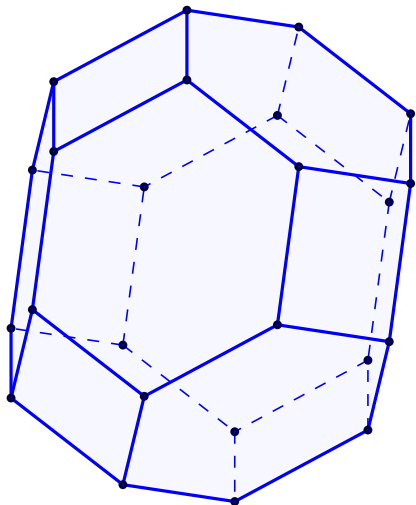
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$P$  and  $P'$  are *equivalent*, denoted  $P \sim P'$ , iff  $P' \trianglelefteq P$  and  $P \trianglelefteq P'$

# Deformations of $\Pi_4$



Permutahedron  $\Pi_4$

Sequence of deformations of  $\Pi_4$

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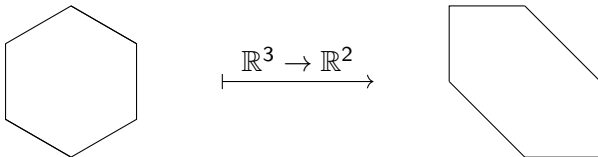
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$\dim \mathbb{SC}_n = 2^n - 1$ ,  $\#_{\text{facets}} \mathbb{SC}_n = \binom{n}{2} 2^{n-2}$ ,  $\#_{\text{rays}} \mathbb{SC}_n = \text{unknown}$

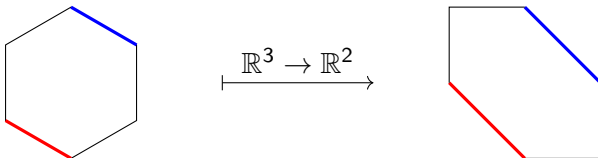
Some faces are “known”: associahedron, graphical zonotopes (arXiv:2111.12422), nestohedra (arXiv:2109.09200), etc.

## *GP-sums and equality sets*

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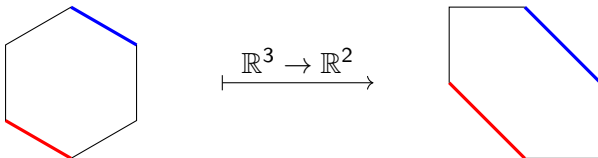


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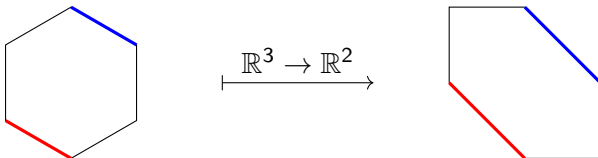
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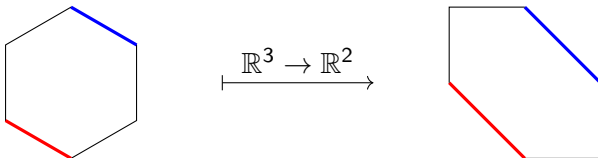
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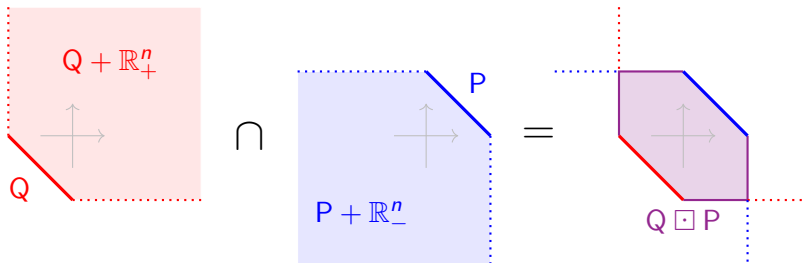
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What's the reciprocal bijection?

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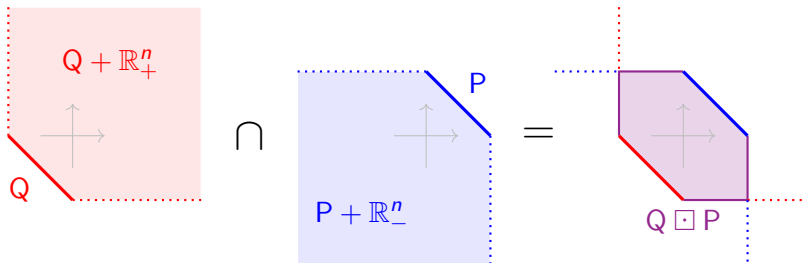
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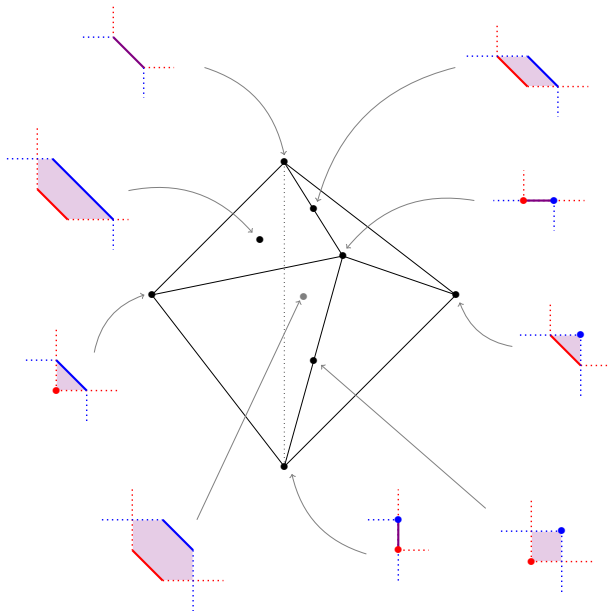
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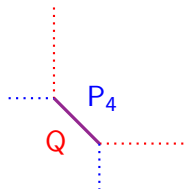
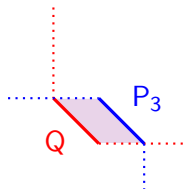
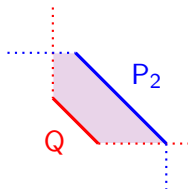
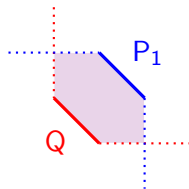
Reciprocal of  $P \mapsto (P^+, P^-)$  is the map  $(P, Q) \mapsto Q \boxdot P$

# $SC_3$ drawn with GP-sums



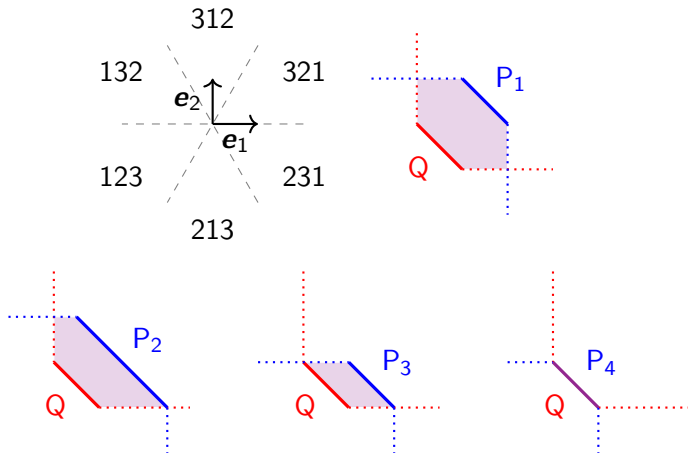
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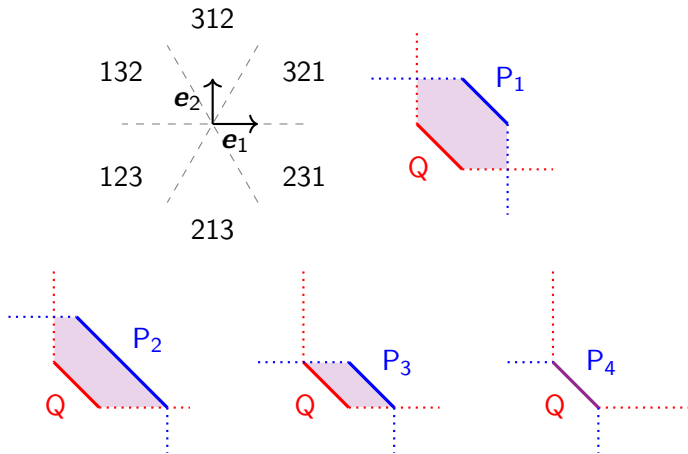
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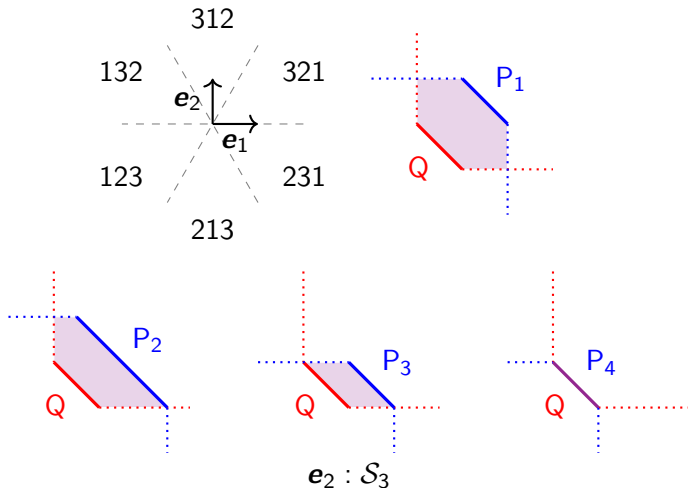
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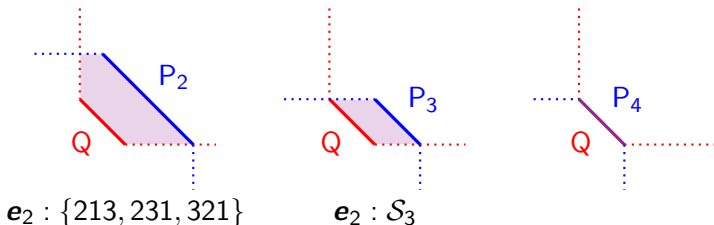
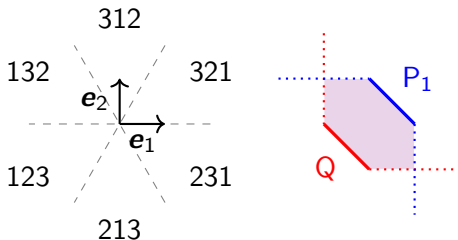
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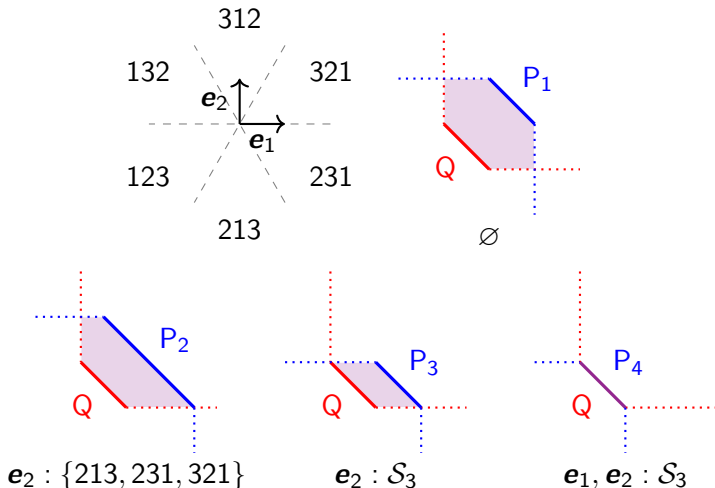
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$$P, Q \in \mathbb{SC}_n$$

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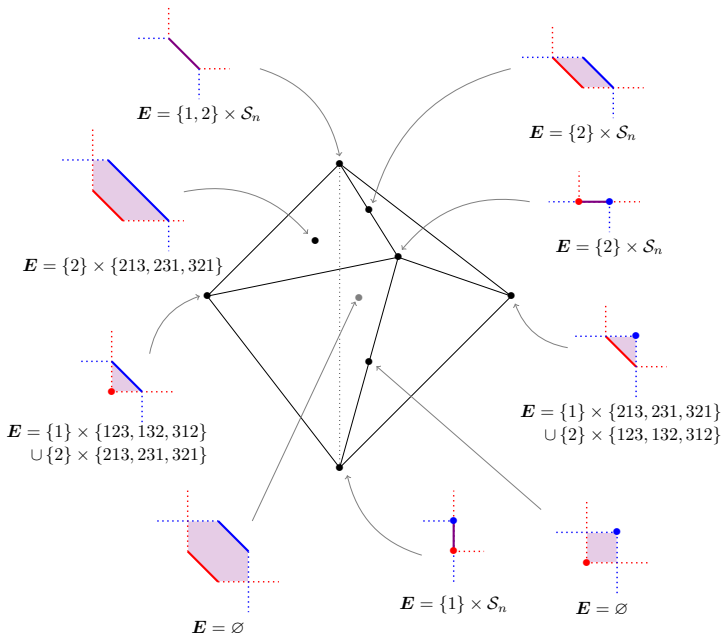
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# $\text{SC}_3$ with equality sets



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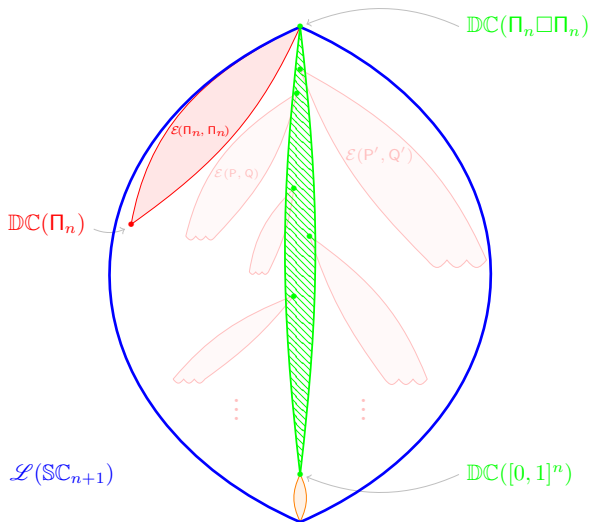
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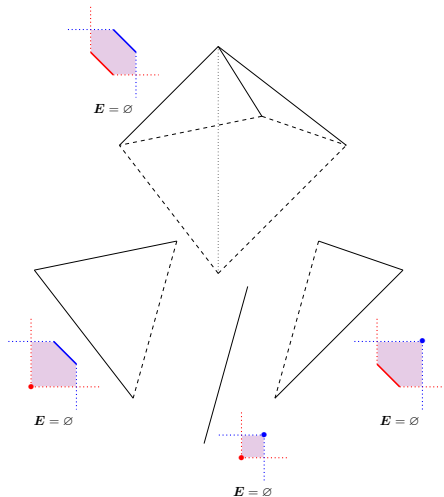
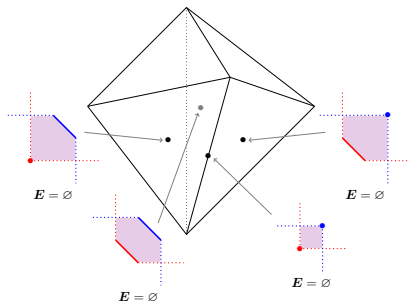
## Corollary (Loho–Padrol–P., 25+)

*The faces of  $\mathbb{SC}_{n+1}$  are in bijection with triples  $(P, Q, \mathbf{E})$  where  $P$  and  $Q$  are taken up to equivalence and  $\mathbf{E} \in \mathcal{E}(P, Q)$ .*

$\mathbb{DC}(\mathbf{P})$ : class of  $\mathbf{P}' \sim \mathbf{P}$ , i.e. faces of  $\mathbb{SC}_n$

$$\mathcal{L}(\mathbb{SC}_n)^2 = \{\mathbb{DC}(\mathbf{Q} \square \mathbf{P}) ; \mathbf{P}, \mathbf{Q} \in \mathbb{SC}_n^2\}$$

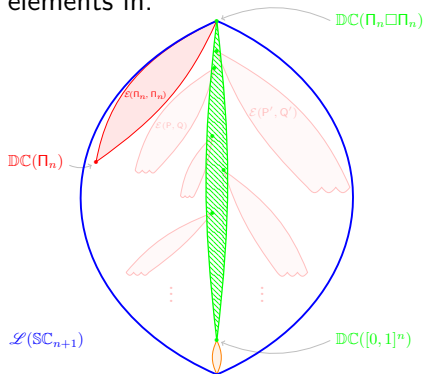




*Applications (now, we can count stuffs)*

# Rays

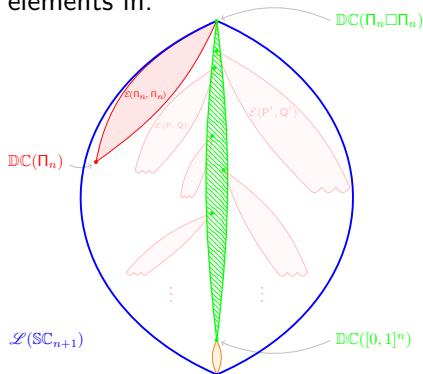
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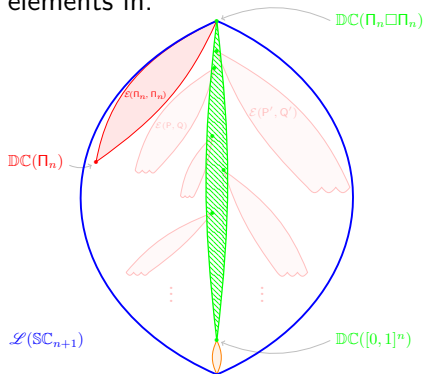


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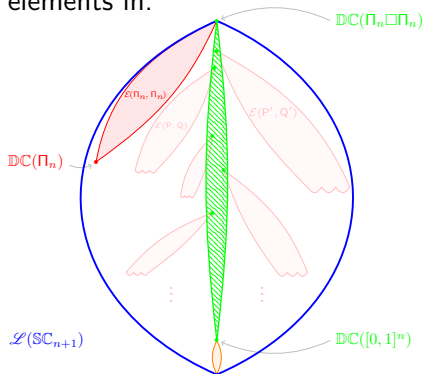


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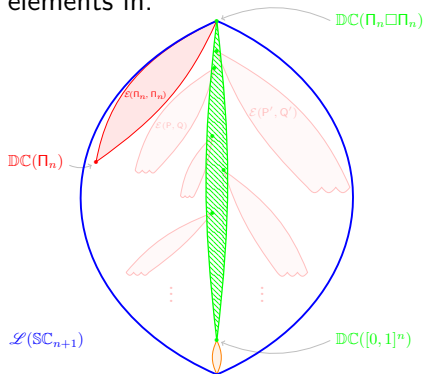


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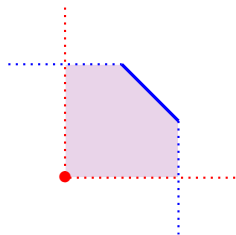
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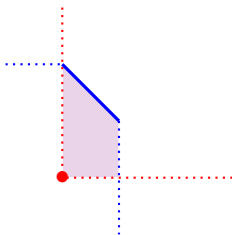
To increase hopes, start with  $P, Q \in \text{rays of } \mathbb{SC}_n$ !

# Rays

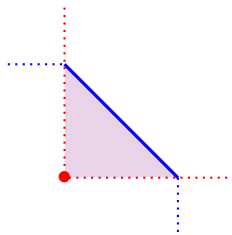
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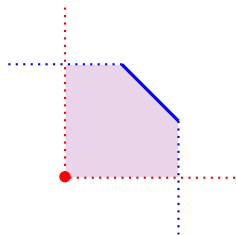
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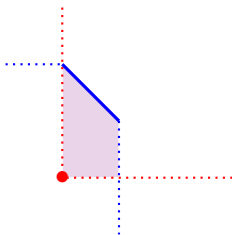
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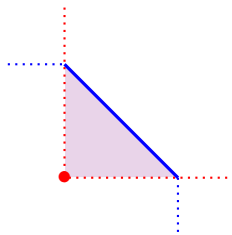
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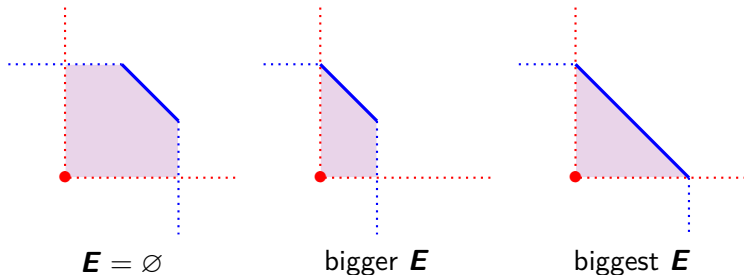


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**Lemma (Loho–Padrol–P., 25+)**

*Technical lemma about non-0 dilations that you wouldn't read.*

**Theorem (Loho–Padrol–P., 25+)**

*If  $P_1, \dots, P_s$  are pair-wise fertile rays of  $\mathbb{SC}_n$ , then you can construct explicitly  $R_1, \dots, R_{s^2}$  pair-wise fertile rays of  $\mathbb{SC}_{n+1}$ .*

# Number of rays of $\mathbb{SC}_n$

$t_n$  := number of rays of  $\mathbb{SC}_n$ .

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$$2^{2^{n-2}} \leq t_n$$

**N.B.:** By Upper Bound Theorem:  $t_n \leq n^{2^n}$ .

Precisely:  $n - 2 \leq \log_2 \log_2 t_n \leq n + \log_2 \log_2 n + 1$

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$$\text{And, } \#_{\text{faces}} \mathbb{SC}_n \geq 22\,108^{2^{n-4}}$$

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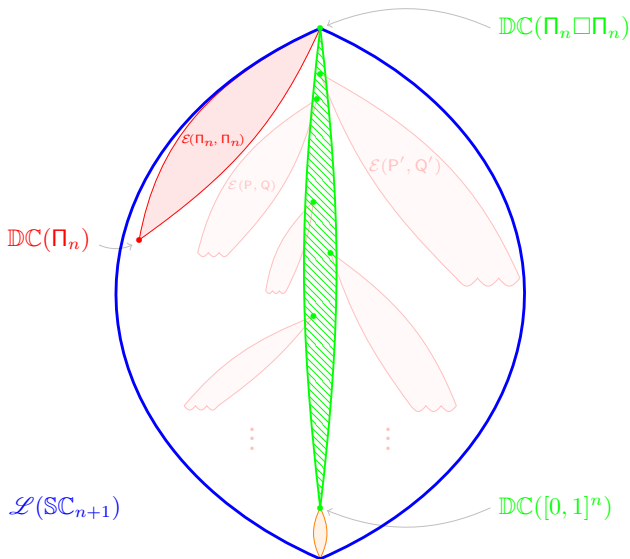
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Corollary (Loho–Padrol–P., 25+)

Number of non-simplicial faces of  $\mathbb{SC}_n \geq \prod_{p < n} \#_{\text{faces}} \mathbb{SC}_p$

In particular,  $\mathbb{SC}_n$  is “very far” from being simplicial, so we should be “very far” for the Upper Bound Theorem equality case.

# Thank you!

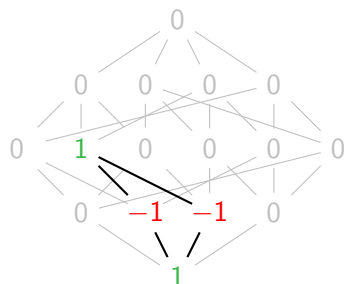
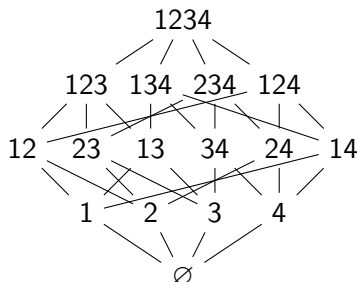


# Tools: submodular dependencies

Notations:  $Sx = S \cup \{x\}$ ,  $(\mathbf{f}_x)_{x \subseteq [n]}$  canonical basis of  $\mathbb{R}^{2^{[n]}}$

## Definition

*Submodular vector*  $\mathbf{n}(S, u, v) = \mathbf{f}_{Suv} - \mathbf{f}_{Su} - \mathbf{f}_{Sv} + \mathbf{f}_S$   
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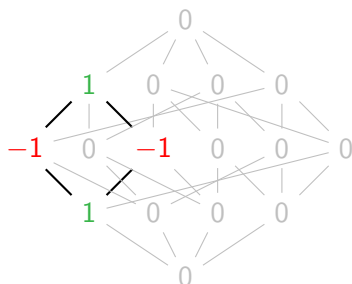
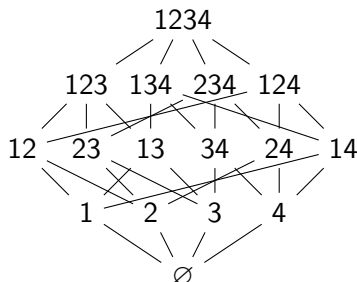


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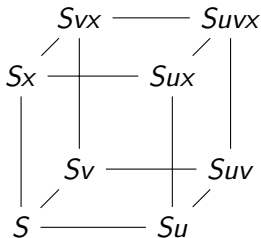
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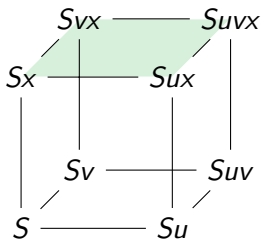
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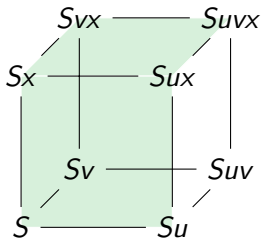
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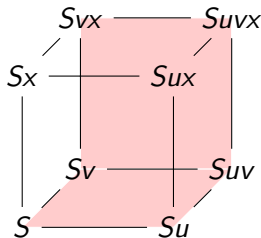
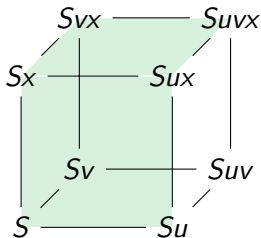
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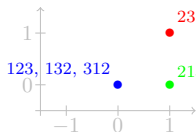
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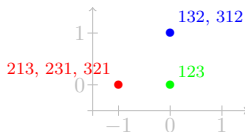
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# Fertility

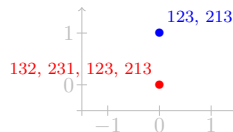
$(P, Q)$  is *fertile* iff there exists  $i \in [n]$  such that for all  $\tau \in S_n$ , either  $P_i^\tau \neq \min_{\sigma \in S_n} P_i^\sigma$  or  $Q_i^\tau \neq \max_{\sigma \in S_n} Q_i^\sigma$



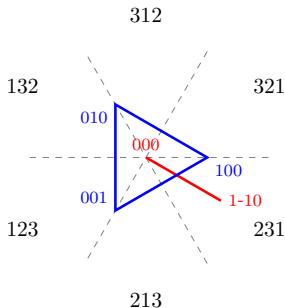
$i = 1$



$i = 2$



$i = 3$



## Multiple choice questions (sometimes several answer possible)

| A sum of 2 co-planar triangles can have                           | 3 edges                          | 4 edges                                 | 5 edges                                 | 6 edges                                   |
|-------------------------------------------------------------------|----------------------------------|-----------------------------------------|-----------------------------------------|-------------------------------------------|
| $\mathcal{N}_{P+Q} = \text{--- of } \mathcal{N}_P, \mathcal{N}_Q$ | union                            | intersection                            | common refinement                       | product                                   |
| Edge directions of $GP \in$                                       | $\{\mathbf{e}_i\}_i$             | $\{\mathbf{e}_i + \mathbf{e}_j\}_{i,j}$ | $\{\mathbf{e}_i - \mathbf{e}_j\}_{i,j}$ | $\{\mathbf{e}_i \pm \mathbf{e}_j\}_{i,j}$ |
| $Z_G$ has no hexagonal face iff $G$ is                            | a tree                           | $K_3$ -free                             | $K_4$ -free                             | complete                                  |
| --- form a basis of $\text{Vect}(\mathbb{SC}_n)$                  | $\{\Delta_X\}_{X \subseteq [n]}$ | nestohedra                              | shard polytopes                         | matroid polytopes                         |
| $\dim \mathbb{SC}_3 =$                                            | 3                                | 4                                       | 5                                       | 11                                        |
| Group of symmetries of $\mathbb{SC}_n$ :                          | $\text{Free}_n$                  | $\mathcal{S}_n$                         | $\mathcal{S}_n \times \mathbb{Z}_2$     | $\mathcal{S}_n \times \mathbb{Z}_n$       |
| # 3-dim indecomposable GP:                                        | 3                                | 5                                       | 7                                       | 37                                        |

1. Show that  $\text{Vert}(P + Q) \subseteq \{\mathbf{u} + \mathbf{v} ; \mathbf{u} \in \text{Vert}(P), \mathbf{v} \in \text{Vert}(Q)\}$ . Find an example with strict inclusion. Prove that  $\mathcal{N}_{P+Q}$  is the common refinement of  $\mathcal{N}_P$  and  $\mathcal{N}_Q$ .
2. Find all the ways to write the 2-dimensional permutahedron as a Minkowski sum of indecomposable polytopes.
3. Prove that  $\mathbf{h}_{P+Q} = \mathbf{h}_P + \mathbf{h}_Q$  and  $\ell_{P+Q} = \ell_P + \ell_Q$ . What are  $\mathbf{h}_{P+t}$  and  $\ell_{P+t}$ ?
4. Prove that a triangle is Minkowski indecomposable. Deduce that all simplicial polytopes are Minkowski indecomposable.

5. Take vectors  $\mathbf{s}, \mathbf{s}', \mathbf{r}_1, \dots, \mathbf{r}_{d-1}$  such that  $C = \text{cone}(\mathbf{s}, \mathbf{r}_1, \dots, \mathbf{r}_{d-1})$  and  $C' = \text{cone}(\mathbf{s}', \mathbf{r}_1, \dots, \mathbf{r}_{d-1})$  are simplicial cones which intersect on their proper face  $C \cap C' = \text{cone}(\mathbf{r}_1, \dots, \mathbf{r}_{d-1})$ . Show that there exist  $\alpha_{\mathbf{s}}, \alpha_{\mathbf{s}'} > 0$  and  $\alpha_{\mathbf{r}_i} \in \mathbb{R}$  such that:

$$\alpha_{\mathbf{s}}\mathbf{s} + \alpha_{\mathbf{s}'}\mathbf{s}' + \sum_i \alpha_{\mathbf{r}_i}\mathbf{r}_i = \mathbf{0}$$

6. Compute (and draw) the deformation cone of a parallelogram (use the edge-lengths point of view). What about a parallelepiped? Show that there is a **unique** way to write a parallelepiped  $P$  as a sum of Minkowski indecomposable polytopes.

7. Show that  $\mathbb{SC}_n$  has  $\binom{n}{2}2^{n-2}$  facets. Give bounds on its number of rays.

8. Show that graphical zonotopes are generalized permutahedra. Show that nestohedra are generalized permutahedra. Show that matroid polytopes are generalized permutahedra. Which are indecomposable?

9. If  $G$  is triangle-free, then  $\mathbb{DC}(Z_G)$  is simplicial (see lecture). Each face of  $\mathbb{DC}(Z_G)$  is associated to a polytope: which polytope?

10. The *weighted graphical zonotope* of a graph  $G = (V, E)$  and weight (on edges)  $\omega : E \rightarrow \mathbb{R}_+^*$  is  $Z_{G,\omega} := \sum_{(i,j) \in E} \omega(i,j)[\mathbf{e}_i, \mathbf{e}_j]$ . Show that:  $\{\text{zonotopes}\} \cap \{\text{generalized permutahedra}\} = \{\text{weighted graphical zonotopes}\}$ . Show that the cone of weighted graphical zonotopes is a section (i.e. intersection with a linear sub-space) of the submodular cone, such that the rays of this section are rays of  $\mathbb{SC}_n$ . What is the dimension of this section?

11. Prove the cubic relations hold. Show that cubic relations generates all linear relations between submodular vectors.